

- 6 (a) (i) $P = VI$ C1
 $60 = 12 \times I$
 $I = 5.0 \text{ A}$ A1 [2]
- (ii) either $V = IR$ or $P = I^2 R$ or $P = V^2 / R$ C1
either $12 = 5 \times R$ or $60 = 5^2 \times R$ or $60 = 12^2 / R$ M1
 $R = 2.4 \Omega$ A0 [2]
- (b) $R = \rho L / A$ C1
 $A = \pi \times (0.4 \times 10^{-3})^2 (= 5.03 \times 10^{-7})$ C1
 $L = (2.4 \times 5.03 \times 10^{-7}) / (1.0 \times 10^{-6})$
 $= 1.2 \text{ m}$ A1 [3]
- (c) resistance is halved M1
either current is doubled or power $\propto 1/R$ M1
power is doubled A1 [3]

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7 (a) nuclei/atoms with same proton number/atomic number

P1